

Experiment Setup: AI-Agent Intervention Study

1. General Design

The study will use a within-subject repeated-measures design with approximately 20 AI-agent personas. Each AI persona will complete both a text-detection task and an image-detection task before and after receiving a short literacy intervention.

The experiment will follow this structure:

1. Pre-intervention text detection
2. Text-detection intervention
3. Post-intervention text detection
4. Pre-intervention image detection
5. Image-detection intervention
6. Post-intervention image detection

All AI personas will complete both the text and image parts of the experiment. This means each persona acts as its own comparison point, allowing pre-intervention and post-intervention performance to be compared within the same agent.

2. Stimuli

Each AI persona will evaluate:

- 10 text stimuli before the text intervention
- 10 text stimuli after the text intervention
- 10 image stimuli before the image intervention
- 10 image stimuli after the image intervention

The pre-intervention and post-intervention stimuli will be different, but they will be designed to be similar in difficulty. All AI personas will receive the same pre-intervention and post-intervention stimulus sets.

For both text and image stimuli, the dataset will use a 50/50 split between AI-generated and human-created content. This means that, within each set of 10 stimuli, 5 will be AI-generated and 5 will be human-created.

3. Intervention

The intervention will consist of written literacy guidance designed to help AI personas identify possible signs of AI-generated content. The intervention will be modality-specific.

For the text-detection task, the AI personas will receive guidance on possible indicators of AI-generated text, such as generic phrasing, overly polished structure, repeated wording patterns, lack of concrete details, predictable introductions or conclusions, and formulaic tone.

For the image-detection task, the AI personas will receive guidance on possible indicators of AI-generated images, such as anatomical implausibilities, stylistic artifacts, functional implausibilities, violations of physics, and social or contextual implausibilities.

The intervention will be given between the pre-intervention and post-intervention tasks. The text intervention will only be shown before the post-intervention text-detection task, and the image intervention will only be shown before the post-intervention image-detection task.

4. Independent Variables

The main independent variable is:

Intervention stage

- Pre-intervention
- Post-intervention

This variable is used to examine whether the literacy intervention improves AI-generated content detection.

The secondary independent variable is:

Modality

- Text
- Image

This variable is used to compare whether the intervention has different effects for text-based content compared to image-based content.

5. Dependent Variables

The study will measure three dependent variables:

1. Detection judgement

For each stimulus, the AI persona will judge whether the content is:

- AI-generated
- Human-created

This will be used to calculate detection accuracy by comparing the persona's judgement with the known ground truth of each stimulus.

2. Confidence

For each judgement, the AI persona will provide a confidence rating on a 1–7 Likert scale.

- 1 = Not confident at all
- 2 = Very low confidence
- 3 = Low confidence
- 4 = Moderate confidence
- 5 = Somewhat high confidence
- 6 = High confidence
- 7 = Very high confidence

This will be used to examine how confident the AI personas are in their detection judgements.

3. Perceived difficulty

For each stimulus, the AI persona will provide a difficulty rating on a 1–7 Likert scale.

- 1 = Very easy
- 2 = Easy
- 3 = Somewhat easy
- 4 = Moderate
- 5 = Somewhat difficult
- 6 = Difficult
- 7 = Very difficult

This will be used to examine how difficult the AI personas perceive the detection task to be.

6. Persona Attributes / Potential Confounding Variables

The personas will be created using a lightweight set of attributes that may plausibly influence detection accuracy, confidence, and perceived difficulty.

The selected persona attributes are:

1. AI literacy

This refers to the persona's understanding of generative AI systems, familiarity with common AI-generated patterns, and ability to recognise indicators of synthetic content.

Options:

- Low: Minimal understanding of generative AI systems. Rarely recognises common AI-generated patterns and has limited awareness of how AI content is produced.
- Medium: Basic understanding of generative AI tools and some familiarity with common indicators of AI-generated content. Can occasionally identify synthetic patterns but may miss subtle cues.
- High: Strong understanding of generative AI systems, training data, and common generation artifacts. Frequently recognises stylistic, structural, or visual indicators associated with AI-generated content.

2. Prior AI use

This refers to the persona's previous use of generative AI tools such as ChatGPT, Claude, Midjourney, DALL-E, or other AI-based systems.

Options:

- None: No meaningful prior interaction with generative AI tools.
- Rare: Very limited use of AI systems, usually for experimentation or occasional curiosity.
- Occasional: Uses AI tools periodically for tasks such as writing assistance, brainstorming, image generation, or information gathering.
- Frequent: Regularly uses AI systems across multiple contexts and is highly familiar with their outputs, strengths, and limitations.

3. Online content skepticism

This refers to how likely the persona is to question online content or suspect that content may be manipulated, misleading, or AI-generated.

Options:

- Low: Generally accepts online content at face value and rarely questions authenticity unless obvious issues are present.
- Moderate: Sometimes questions the reliability or authenticity of online content, especially when inconsistencies or unusual details are noticeable.
- High: Frequently evaluates online content critically and actively looks for signs of manipulation, misinformation, or synthetic generation.

4. Attention to detail

This refers to how carefully the persona tends to inspect visual or textual details before making a judgement.

Options:

- Low: Tends to rely on quick impressions and may overlook subtle inconsistencies or fine details.
- Medium: Pays attention to noticeable details but may not consistently inspect all aspects of the content carefully.
- High: Carefully analyses wording, structure, visual elements, and contextual consistency before making a judgement.

5. Prior exposure to misinformation

This refers to whether the persona has previously encountered misinformation, fake news, scams, misleading online content, or discussions about manipulated media.

Options:

- Low: Limited prior exposure to misinformation or manipulated media discussions.
- Medium: Some prior exposure to misinformation, fake news, or online discussions about deceptive content.
- High: Extensive prior exposure to misinformation, manipulated media, online scams, or educational discussions about deceptive digital content.

These attributes are not the main independent variables of the experiment. Instead, they are treated as persona-level characteristics that may influence the dependent variables. They may help explain variation in detection accuracy, confidence, and perceived difficulty across AI personas.

7. Capturing Responses from the API

The API response structure will depend on the type of AI-agent setup used.

Scenario A: Personally controlled ChatGPT-based AI agents

If the AI personas are created and run manually through a controlled ChatGPT setup, the input prompt and persona attributes will already be visible to the researcher. In this case, the response after each stimulus should include:

- Stimulus ID
- Detection judgement
- Confidence rating
- Difficulty rating
- Short reasoning

Response format:

```
{
  "stimulus_id": "T01",
  "modality": "text",
  "stage": "pre_intervention",
  "judgement": "AI-generated",
  "confidence": 5,
  "difficulty": 4,
  "reasoning": "The text feels polished and somewhat generic, with broad phrasing and limited concrete details."
}
```

Scenario B: External or less transparent AI-agent system

If the AI-agent system does not make the input prompt or persona attributes directly visible, then the response should also include the persona attributes. This would make it easier to connect each response to the specific characteristics of the AI persona.

Response format:

```
{
  "persona_id": "P01",
  "ai_literacy": "Medium",
  "prior_ai_use": "Occasional",
  "online_content_skepticism": "Moderate",
  "attention_to_detail": "High",
  "prior_exposure_to_misinformation": "Medium",
  "stimulus_id": "T01",
  "modality": "text",
  "stage": "pre_intervention",
  "judgement": "AI-generated",
  "confidence": 5,
  "difficulty": 4,
  "reasoning": "The text feels polished and somewhat generic, with broad phrasing and limited concrete details."
}
```

For analysis, the most important fields are the judgement, confidence rating, and difficulty rating. The reasoning field is useful for qualitative interpretation, but the main quantitative analysis will be based on detection accuracy, confidence, and difficulty.

8. Accuracy Calculation

Detection accuracy will be calculated by comparing each persona's judgement with the known ground truth of the stimulus.

For each persona, accuracy can be calculated separately for:

- Pre-intervention text detection
- Post-intervention text detection
- Pre-intervention image detection
- Post-intervention image detection

For example:

```
Text pre-intervention accuracy = number of correct pre-text judgements / 10
Text post-intervention accuracy = number of correct post-text judgements / 10
Image pre-intervention accuracy = number of correct pre-image judgements / 10
Image post-intervention accuracy = number of correct post-image judgements / 10
```

Improvement scores can then be calculated:

Text improvement = post-text accuracy - pre-text accuracy
Image improvement = post-image accuracy - pre-image accuracy

These improvement scores can be used to compare whether the intervention was more effective for text or image detection.